

Qatar-2b

R-0744

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_250201 · created 2026-07-28T02:11:54+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460707.9837 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.228 +/- 0.066
Transit depth (Rp/Rs)^2	5.19 +/- 3.0 [%]
Orbital Inclination (inc)	89.9 +/- 1.74
Airmass coefficient 1 (a1)	85.34 +/- 66.95
Airmass coefficient 2 (a2)	-2.98 +/- 0.99
Scatter in the residuals of the lightcurve fit is	76.48 %
Best Comparison Star	None
Optimal Aperture	1.58
Optimal Annulus	5.0
Transit Duration (day)	0.0531 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.228 $\pm$ 0.066 vs 0.16327 (deviation 0.85 σ)
Duration fit vs published	0.0531 d vs 0.0767 d (deviation 7.07 σ)
Mid-time O-C	-0.7 min
Depth SNR	3.0
Residual scatter	76.48 %

Light curve

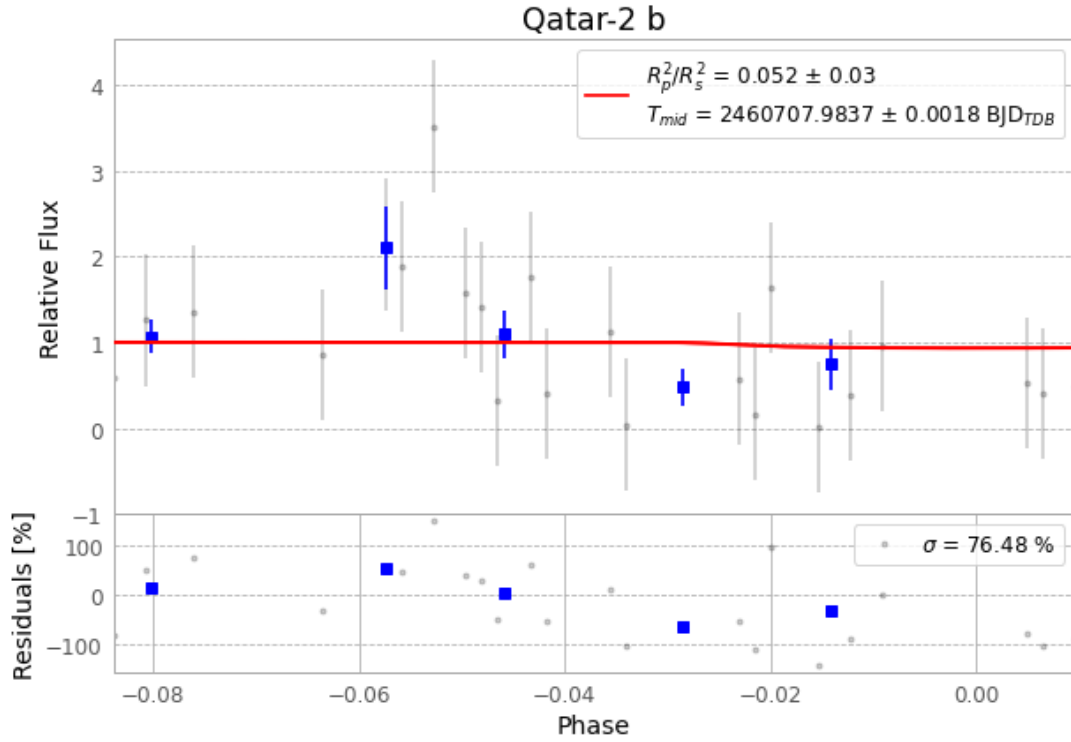


Figure 1 — FinalLightCurve_Qatar-2 b_2025-02-01.png (SHA-256 5596cfd80684aec9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [497, 297], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ee89f4474d45cc32...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

67 FITS frames (44 MB), Qatar-2250201083618.FITS → Qatar-2250201115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-250201.log (4b2f1b73ce71...), FinalLightCurve_Qatar-2 b_2025-02-01.png (5596cfd80684...), FinalLightCurve_Qatar-2 b_2025-02-01.csv (803c93b4941d...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 02:11Z.